Thesis Report – Graduation Internship

Quantum Imaging with Undetected Photons: Enhanced Image processing & User Interface

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Foreword

As a student at Saxion – University of Applied Sciences, this project will be a graduation internship that serves as the final step in completing the Bachelor of Applied Computer Science program.

Quantum Photonics is an emerging field that explores the quantum properties of light and their applications in various technologies, including quantum imaging. Quantum Imaging with Undetected Photons (QIUP) is an advanced technique that allows for imaging objects without directly detecting the photons that interact with them, leveraging quantum entanglement and interference effects.

The Applied Nanotechnology (ANT) research group at Saxion is actively involved in advancing quantum photonics technologies, including QIUP. This project will focus on enhancing the image processing and developing a user-friendly interface for the QIUP system, aiming to improve the quality of the images produced and make the technology more accessible to researchers and practitioners in the field.

Summary

The aim of this report is to make Quantum technologies more accessible to a wider audience by contributing to the advancement of quantum imaging technology. Fast Fourier Transform (FFT), an advanced image processing method, was used to enhance the quality of images produced by QIUP. Additionally, a user-friendly interface was developed to allow researchers and practitioners to easily interact with the QIUP system, facilitating the adoption and application of this cutting-edge technology in various fields.

— TO DO —: Results, Conclusions, and Recommendations.

Contents

Abbreviations

QIUP Quantum Imaging with Undetected Photons

ANT Applied Nanotechnology

FFT Fast Fourier Transform

TLC Talent & Learning Center

PPLN Periodically Poled Lithium Niobate

CMOS Complementary Metal-Oxide-Semiconductor

SPDC Spontaneous Parametric Down-Conversion

CCD Charged Coupled Device

GUI Graphical User Interface

KPZ101 KCube Piezo Controller

KSG101 KCube Strain Gauge Meter

SDK Software Development Kit

TSI Thorlabs Scientific Imaging

CLR Common Language Runtime

FFTW Fastest Fourier Transform in the West

SIMD Single Instruction, Multiple Data

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1 Introduction

This chapter provides an introduction to the project, explains the background and context of the assignment and also describes the setup of the demonstrational platform for QIUP at the Talent & Learning Center (TLC) lab within the ANT research group.

1.1 Background of the Assignment

As part of the Quantum Delta program, this project aims to accelerate the development and adoption of quantum technologies [1]. A primary challenge in this field is the non-intuitive nature of quantum properties such as entanglement and indistinguishability, which is being leveraged in QIUP to enable imaging without directly interacting with the photon that detects the sample.

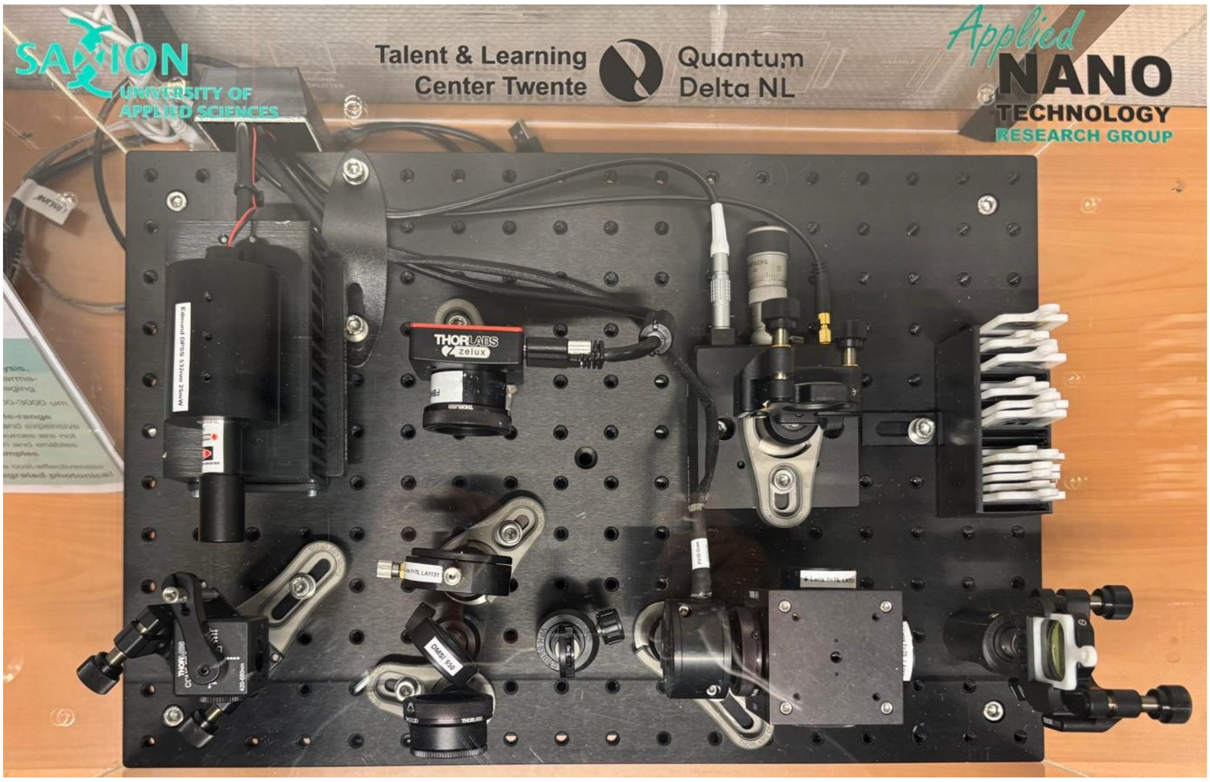
The demonstrational setup for QIUP at the TLC lab within the ANT research group provides a platform for exploring and enhancing this technology.

1.2 The Setup

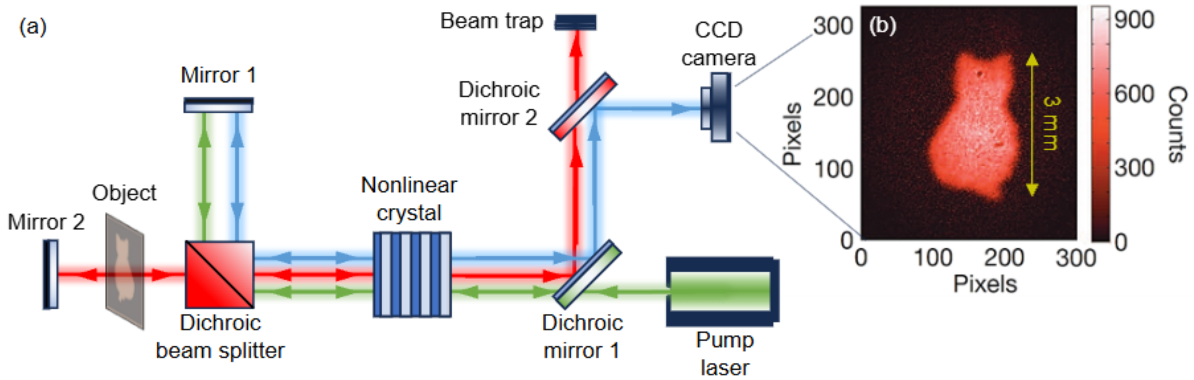
The experimental implementation of the quantum imager is shown in Figure 1. The system consists of:

- Continuous-wave pump laser
- Periodically Poled Lithium Niobate (PPLN) nonlinear crystal
- Dichroic beam splitter
- Piezoelectric-stage connected with a mirror
- Thorlabs Zelux Complementary Metal-Oxide-Semiconductor (CMOS) camera

Inside the PPLN crystal, Spontaneous Parametric Down-Conversion (SPDC) splits pump photons from a continuous-wave laser with frequency of 532 nm into signal (visible) photons with a frequency of 810 nm and idler (infrared) photons with a frequency of 1550 nm (telecommunication). The idler interacts with the object, while the signal does not. Upon recombination, interference occurs and is recorded by the CMOS camera.



(a) Top-Down view on demonstrational setup at Saxion.



(b) Schematic of demonstrational setups functionality.

Figure 1: Demonstrational Setup and Schematic.

The system relies on the principle of induced coherence, utilizing a nonlinear interferometer.

As detailed in the schematic (Figure 1b), a continuous wave pump laser is directed into a nonlinear crystal specifically, a PPLN crystal. Through SPDC, the pump photon splits into a signal photon (visible spectrum) and an idler photon (infrared). A dichroic beam splitter separates these paths: the visible signal is reflected by a reference mirror (Mirror 1), while the infrared idler interacts with the target object before being reflected by Mirror 2. The idler photons that pass through the object contain spatial information about its absorption or phase profile. When the idler and signal beams recombine at the crystal, quantum interference modulates the intensity of the signal beam based on the idler's path information. Consequently, the image of the object which was only "seen" by the infrared light is transferred to the visible spectrum and detected by the Charged Coupled Device (CCD) camera. The physical realization of this schematic is shown in Figure 1a. The setup is constructed on an optical breadboard to ensure stability. The pump source directs the beam through conditioning optics towards the central crystal mount. The detection module utilizes a Thorlabs Zelux CMOS camera to capture the interference patterns.

1.3 Problem Definition and Objectives

Problem Definition

The experimental setup for QIUP serves as a powerful proof-of-concept for mid-infrared imaging. However, the current setup has limitations in terms of image quality and user accessibility. There are two main software tools by Thorlabs that are used to control the setup and process the images, Thorcam [2] (left part of the figure 2) and Kinesis [3] (right part of the figure 2), which are used for camera and piezoelectric actuator control, but they are not optimized for QIUP applications or any image processing. Using this approach is rather unintuitive and requires a lot of manual work to only get gray scale images, which do not have the best quality as illustrated in Figure 2.

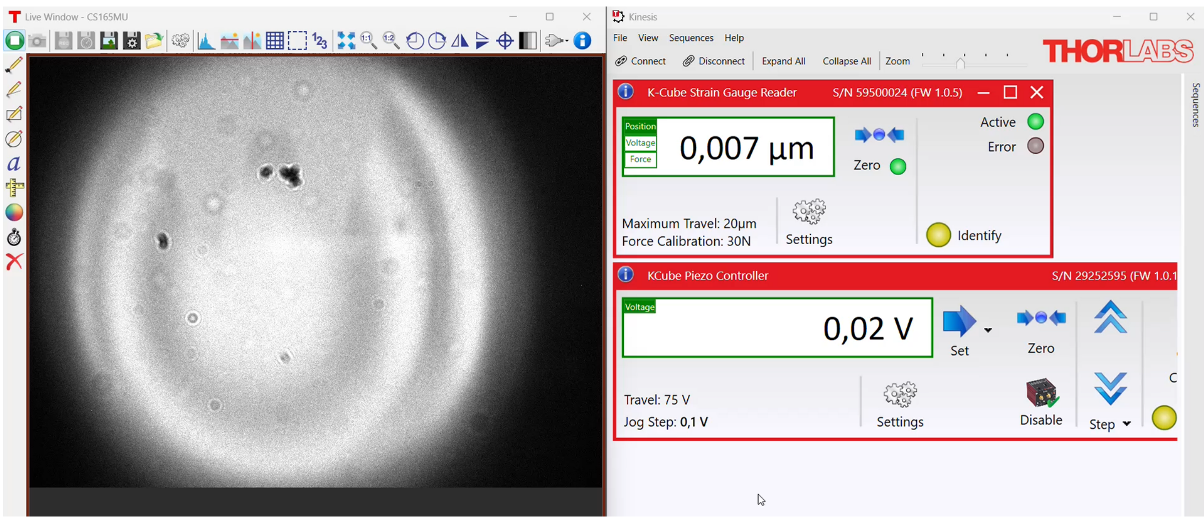


Figure 2: Old Showcase of the QIUP Setup.

Objectives

The main objective of this project is to transform the current laboratory setup into a robust, automated, and user-friendly demonstrator platform for QIUP technology demonstration, which can be divided into the following sub-objectives:

1. **Advanced Image Processing:** Captured images will be processed using advanced techniques, such as FFT, to enhance image quality and extract more detailed information like visibility, contrast and phase from the raw data.

2. **Hardware Automation System:** The current use of multiple software tools for controlling the hardware components will be streamlined into a single, cohesive automation system that can manage all aspects of the QIUP setup, including camera control, piezoelectric actuators, data acquisition and image processing.

3. **User Interface Development:** A user-friendly interface will be developed to allow researchers and practitioners to easily interact with the QIUP system, facilitating the adoption and application of this cutting-edge technology in various fields. The interface will provide intuitive controls for operating the hardware, visualizing the processed images, and accessing the key metrics visibility and contrast phase.

1.4 Scope of Work

To consider this project successful, the expected technical results have been categorized using the MoSCoW method:

Must Have:

- Hardware Automation System: A robust Python-based control script capable of communicating with and reliably driving both the Thorlabs CMOS camera and the piezoelectric stage.

- Image Processing Core: A functional FFT-based algorithm capable of near real-time reconstruction of visibility, contrast, and phase images from the raw sensor data.

Should Have:

- Functional Graphical User Interface (GUI): An intuitive graphical user interface that allows the operator to control the hardware, acquire live images, and display the reconstructed quantum interference patterns.

- Performance Optimization: Validated and optimized code execution that ensures the imaging performance is fast and smooth enough for live audience demonstrations.

- Advanced camera settings: The ability to adjust camera settings such as exposure time, gain, and frame rate through the software interface to optimize image quality under different conditions.

Could Have:

- Advanced GUI Capabilities: Additional features in the software, such as one-click calibration presets, advanced data visualization tweaks, or the ability to export/upload acquired data.
- New Samples: For demonsstration purposes, new samples (transparent or opaque) with creative visualizations could be introduced to better showcase the capabilities of QIUP technology .

Won't Have:

- Hardware Upgrades: The physical integration of new hardware components not present in the current setup.

1.5 Approach & Methodology

This project combines software development with experimental physics validation. To manage the iterative nature of development alongside research and hardware integration, the project utilizes the Scrum framework. This allows for flexibility while ensuring continuous delivery of features and frequent validation against the physical setup.

Development Process

The project is divided into sprints lasting 1–4 weeks. This duration provides a balance between sufficient development time and frequent feedback loops. At the end of each sprint, the results are reviewed and the subsequent sprint is adjusted based on the progress of the QIUP system integration.

System Architecture and Tools

The software is developed as a modular system using the following tools:

- **Language & IDE:** Python is used for the entire system development within the Visual Studio Code environment.

- **Hardware Control:** Integration with the Thorlabs piezoelectric stage with corresponding Strain Gauge Meter (KPZ101/KSG101) and CMOS camera is handled via the Kinesis .NET API and ThorCam SDK.
- **Data Management:** Project assets are saved in a secure directory on the ANT internal network, ensuring compliance, data integrity, and unified access.

FFT-Based Image Processing

For a QIUP demonstrator, the most important information is not found in a single static image, but rather in how the image changes. Because the system utilizes an interferometer with a scanning piezoelectric stage, we expect a very specific physical effect: as the optical path length changes, the light intensity at any given point will oscillate sinusoidally, creating moving interference fringes. [4]

To capture and translate these interferometric effects into a clear image, this project implements a pixel-wise FFT approach. Rather than analyzing the spatial patterns of a single frame, the software treats every individual pixel on the camera as an independent sensor. It tracks how the brightness of each pixel changes over a sequence of phase-shifted frames, creating a time-domain signal.

1.6 Outline of the Report

1. **Introduction:** Provides background on quantum imaging technology, defines the problem statement, outlines project objectives, and presents the methodology and scope of work.
2. **Analysis / Specifications:** Details the requirements analysis, system specifications, and constraints for the enhanced image processing and user interface development.
3. **Functional Design:** Describes the functional architecture, user workflows, and feature specifications for the image processing algorithms and UI components.
4. **Technical Design:** Presents the technical implementation approach, software architecture, technology stack, and design patterns used for the project.
5. **Realisation:** Documents the actual implementation of the system, including code development, integration, and deployment of the quantum imaging software.
6. **Testing:** Covers testing strategies, test cases, results, and validation of the image processing quality and user interface functionality.

7. **Conclusions & Recommendations:** Summarizes key findings, evaluates achievement of objectives, and provides recommendations for future improvements and research directions.

2 Analysis / Specifications

In this chapter, the requirements for the QIUP project will be analyzed. Additionally to the requirements, the constraints will be discussed and specified

2.1 Physical Specifications

To grasp how this demonstrational setup works, it is crucial to understand the underlying physical principles and components that enable QIUP technology.

Nonlinear Interferometry
To understand how QIUP works, it is essential to first understand whats behind it: the nonlinear interferometer. In classical optics, a standard interferometer uses mirrors and beam splitters to divide a single beam of light, send the halves along different paths, and recombine them to measure interference patterns. A nonlinear interferometer, however, replaces standard beam splitters with nonlinear crystals to exploit the quantum properties of light [5].

Spontaneous Parametric Down-Conversion
The core of this system is a quantum optical process called SPDC [6]. When a high-energy laser (the laser pump) illuminates a specialized nonlinear crystal in our case the PPLN crystal, a photon will spontaneously split into a pair of photons at different wavelengths [5]. These two newly created photons are called the signal photon and the idler photon. In the context of this demonstrator, the signal photon is in the visible spectrum (detected by the CMOS camera), while the idler photon is in the infrared spectrum (which will interact with the object).

Induced Coherence and "Which-Way" Information
The foundational principle behind QIUP is a phenomenon known as "induced coherence without induced emission" [5]. Imagine an optical setup configured so that a signal-idler photon pair could be created in a first crystal or a second identical crystal. The system is precisely aligned so that the path of the idler photon from the first crystal overlaps perfectly with the emission path of the idler from the second crystal, which in our case is the same crystal where the original light is reflected back to.

Indistinguishability and Interference
Quantum interference is observed at the camera only if the path taken by the photon is fundamentally unknown [5, 6]. In QIUP, the alternating fringe pattern extracted via FFT exists because the signal photon's origin is undefined. By perfectly overlapping the infrared idler and signal beams, the setup intentionally erases the "which-way" (or welcher-weg) information regarding where the photon pair was generated [6, 5]. This forced spatial indistinguishability induces coherence between the visible signal photons,

enabling them to interfere even though they never interacted with the sample [7, 5].

Extracting the Image
To turn this quantum phenomenon into an imaging system, the sample is placed in the path of the idler photon. The transmission and phase properties of the object directly control the visibility of the interference fringes seen in the signal beam [5]. The infrared idler photon gathers the physical information about the object, but it is entirely discarded and never reaches a detector [6]. Meanwhile, the visible signal photon never interacts with the object, yet its interference pattern, captured by the CMOS camera and processed via FFT, contains the complete image [8, 7].

2.2 Hardware Specifications

The demonstrational setup has several hardware components that are essential for the operation of the QIUP system.

Piezoelectric Stage
To utlilize the principle of induced coherence, the setup relies on a piezoelectric stage to precisely control the optical path length of the signal beam in µm. This is necessary to create the phase shifts required for the interference pattern to emerge and be captured by the camera.

The piezoelectric stage consists of two components:

- Thorlabs KPZ101

Figure 3: Thorlabs KPZ101 Piezoelectric Controller.

With the KPZ101 we are able to control the position of the piezoelectric stage with high precision, allowing for accurate phase scanning and modulation of the interference pattern. This is possible by applying voltage to the piezoelectric actuator, which causes it to expand or contract, thus changing the optical path length of the signal beam. [9] The KPZ101 is visualized in Figure 3.

- Thorlabs KSG101

Figure 4: Thorlabs KSG101 Strain Gauge Meter.

The KSG101 is a strain gauge meter that provides feedback on the position of the piezoelectric stage, ensuring precise control and stability during the phase scanning process, which is needed because we need to have exactly one phase cycle otherwise we encounter spectral leakage [10]. The KSG101 is visualized in Figure 4.

Open & Closed loop mode

The piezoelectric stage can operate in both open-loop and closed-loop modes. In open-loop mode, the position of the stage is controlled solely by the input voltage, without any feedback, which leads to less precise control because the displacement for a given voltage can vary since the piezoelectric material's response is not linear [9]. In closed-loop mode, the system uses feedback from the KSG101 to adjust the voltage and maintain a precise position, which is crucial for achieving stable and accurate phase modulation in the QIUP setup [10].

CMOS Camera

The Thorlabs Zelux CMOS camera is used as the detector in the QIUP setup to capture signal photons, which contain the image information retrieved by the idler photons.

Figure 5: Thorlabs Zelux CMOS Camera.

2.3 Software Specifications

To control the hardware components and process the captured images, several software tools and technologies are utilized in this project.

Programming Language
Python is chosen as the programming language of the application as it provides a wide range of optimized libraries for image processing, hardware control and GUI development, making it an ideal choice for this project. Additionally, as Python is widely used in scientific computing and research, it allows for easier collaboration and integration with existing tools and workflows. Furthermore, Python's simplicity and readability enable further development for researchers who may not have extensive programming experience, which is important for the accessibility of the QIUP technology.

Software Development Kits
To integrate hardware control seamlessly within the Python environment, two primary Software Development Kit (SDK) from Thorlabs are utilized. First, the Thorlabs Scientific Imaging (TSI) Camera Python SDK serves as the interface for the Zelux CMOS camera. This SDK acts as a Python-native wrapper around Thorlabs' core C/C++ imaging libraries, granting direct programmatic access to camera parameters such as exposure time and gain. Most importantly, the TSI SDK allows the continuous stream of raw sensor data to be ingested directly into Python memory as multidimensional arrays like numpy arrays. This zero-copy or low-overhead transfer is an essential prerequisite for the low-latency, pixel-wise FFT processing required to reconstruct the quantum images in near real-time [11].

Second, the precision motion control of the interferometer's piezoelectric stage relies on the Thorlabs Kinesis .NET API. Because the Kinesis framework is inherently designed for Windows .NET environments, a bridge module such as `pythonnet` (Python for .NET) is used to load the Common Language Runtime (CLR) directly into the Python backend. This integration is crucial for integrating both the KPZ101 Piezo Controller and the KSG101 Strain Gauge Reader. Through the Kinesis API, the software can actively monitor the real-time physical displacement feedback from the strain gauge and issue controlled, closed-loop voltage commands to the piezo driver [12]. Establishing this programmatic link enables precise, µm synchronization between the camera's frame acquisition and the piezo stage's phase-stepping ensuring that exactly one phase cycle is captured per measurement sequence to prevent spectral leakage during FFT analysis.

Libraries
Beyond the hardware-specific SDKs, the software architecture relies on a stack of highly optimized open-source Python libraries to achieve the project's data processing and visualization objectives.

For data handling, NumPy is utilized as the foundational package for scientific computing, providing the high-performance multidimensional array objects necessary for storing and manipulating the raw camera frames.

To execute the core image processing algorithm, the PyFFTW library [13] is used. PyFFTW is a Pythonic wrapper around the Fastest Fourier Transform in the West (FFTW) C library [14, 15]. Because the demonstrator aims for near real-time reconstruction of the quantum images during live presentations, standard Python-based FFT implementations are computationally too slow. PyFFTW solves this by utilizing multi-threading and CPU-specific Single Instruction, Multiple Data (SIMD) instructions, ensuring the FFT remains smooth and responsive.

Finally, for the development of the user-friendly interface, a primary objective of this project, the graphical framework PyQT5 is used. This allows for the creation of an intuitive dashboard where operators can seamlessly view live camera feeds, adjust piezo-stage parameters, and observe the reconstructed visibility, contrast, and phase maps in one unified environment.

2.4 Requirements

User Requirements

User requirements define what the operators (e.g., researchers, students, or presenters) need to achieve when interacting with the QIUP demonstrator platform.

ID	**Requirement Description**	**Priority**
UR-01	The user must be able to view live, reconstructed visibility, contrast, and phase images through a single GUI.	Must
UR-02	The user must be able to initiate an automated phase-scanning sequence with a single interaction.	Must
UR-03	The user should be able to adjust camera parameters (e.g., exposure time, gain) directly from the GUI.	Should
UR-04	The user should experience smooth, near real-time image updates suitable for live audience demonstrations.	Should
UR-05	The user could have the ability to save or export the processed data and configuration presets for future experiments.	Could

Table 1: User Requirements for the QIUP setup.

System Requirements

System requirements detail the technical, functional, and performance specifications that the software and hardware integration must fulfill to satisfy the user requirements.

ID	**Requirement Description**	**Priority**
SR-01	The system must establish a programmatic connection to the Thorlabs Zelux CMOS camera using the Thorlabs TSI Python SDK.	Must
SR-02	The system must control the KPZ101 Piezo Controller and read feedback from the KSG101 Strain Gauge via the Kinesis .NET API in closed-loop mode.	Must
SR-03	The system must synchronize camera frame acquisition with the piezo stage stepping to capture exactly one phase cycle, preventing spectral leakage.	Must
SR-04	The system must ingest raw camera frames as image stacks and perform pixel-wise FFT analysis to extract the interference fringe parameters.	Must
SR-05	The system should utilize the PyFFTW library to leverage multi-threading and SIMD instructions, ensuring low-latency image processing.	Should

Table 2: System Requirements for the QIUP setup.

3 Functional Design

4 Technical Design

5 Realisation

6 Testing

7 Conclusions and Recommendations

7.1 Conclusions

7.2 Recommendations

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Appendices